

Runner	HPC	Container	Test	Status	Time (s)
cwltool	laptop	docker	workflow-base	OK	5.0
cwltool	laptop	docker	workflow-req	OK	4.0
cwltool	laptop	none	workflow-base	OK	3.0
cwltool	laptop	none	workflow-req	OK	3.0
cwltool	laptop	singularity	workflow-base	OK	4.0
cwltool	laptop	singularity	workflow-req	1	3.0
cwltool	lumi	none	workflow-base	1	87.0
cwltool	lumi	none	workflow-req	OK	4.0
cwltool	lumi	singularity	workflow-base	OK	79.0
cwltool	lumi	singularity	workflow-req	1	5.0
cwltool	mn5	none	workflow-base	OK	8.0
cwltool	mn5	none	workflow-req	OK	3.0
cwltool	mn5	singularity	workflow-base	1	3.0
cwltool	mn5	singularity	workflow-req	1	2.0
streamflow	laptop	docker	workflow-base	OK	3.0
streamflow	laptop	docker	workflow-req	1	1.0
streamflow	laptop	none	workflow-base	OK	2.0
streamflow	laptop	none	workflow-req	1	2.0
streamflow	laptop	singularity	workflow-base	timeout	31.0
streamflow	laptop	singularity	workflow-req	1	1.0
streamflow	lumi	none	workflow-base	1	113.0
streamflow	lumi	none	workflow-req	1	3.0
streamflow	lumi	singularity	workflow-base	timeout	300.0
streamflow	lumi	singularity	workflow-req	1	3.0
streamflow	mn5	none	workflow-base	1	16.0
streamflow	mn5	none	workflow-req	1	1.0
streamflow	mn5	singularity	workflow-base	1	1.0
streamflow	mn5	singularity	workflow-req	1	1.0
toil	laptop	docker	workflow-base	OK	18.0
toil	laptop	docker	workflow-req	1	4.0
toil	laptop	none	workflow-base	OK	11.0
toil	laptop	none	workflow-req	OK	11.0
toil	laptop	singularity	workflow-base	OK	14.0
toil	laptop	singularity	workflow-req	1	14.0
toil	lumi	none	workflow-base	1	124.0
toil	lumi	none	workflow-req	OK	14.0
toil	lumi	singularity	workflow-base	OK	113.0
toil	lumi	singularity	workflow-req	1	20.0
toil	mn5	none	workflow-base	1	18.0
toil	mn5	none	workflow-req	1	8.0
toil	mn5	singularity	workflow-base	1	10.0
toil	mn5	singularity	workflow-req	1	9.0

Table 1: HPC workflow execution results across runners, platforms, and container backends.